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Questions

Q1.

Two students observe a book of mass 0.2 kg fall vertically from rest from a shelf that is 1.5 m above the floor.

Student *A* suggests that the book is modelled as a particle falling freely under gravity.

(a) Use student *A*'s model to find the time taken for the book to reach the floor.

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(3)

Student *B* suggests an improved model where the book is modelled as a particle experiencing a constant resistance to motion of magnitude *R* newtons.

Given that the time taken for the book to reach the floor is 0.6 seconds,

(b) use student *B*'s model to find the value of *R*

(5)

(Total for question = 8 marks)

(**Q03 WME01/01, June 2023**)

Q2.

A motorbike is moving with constant acceleration along a straight horizontal road.

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The motorbike passes a point *P* and 10 seconds later passes a point *Q*.

The speed of the motorbike as it passes *Q* is 28 m s^{-1}

Given that *PQ* = 220 m,

(a) find the acceleration of the motorbike,

(3)

(b) find the distance travelled by the motorbike during the fifth second after passing *P*

(4)

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(Total for question = 7 marks)

(Q02 WME01/01, June 2022)

Q3.

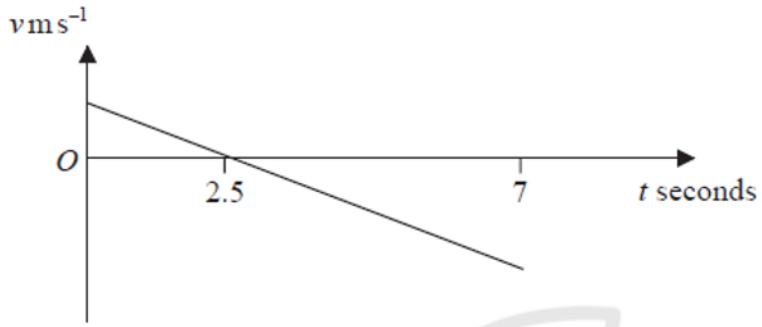


Figure 3

A small ball is thrown vertically upwards at time $t = 0$ from a point A which is above horizontal ground. The ball hits the ground 7 s later.

The ball is modelled as a particle moving freely under gravity.

The velocity-time graph shown in Figure 3 represents the motion of the ball for $0 \leq t \leq 7$

(a) Find the speed with which the ball is thrown.

(2)

(b) Find the height of A above the ground.

(3)

(Total for question = 5 marks)

(Q06 WME01/01, Oct 2021)

Q4.

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A van travels with constant acceleration along a straight horizontal road.

The van passes a point A with speed u m s^{-1} and 20 seconds later passes a point B with speed 28 m s^{-1}

The distance AB is 400 m.

(a) Show that $u = 12$

(2)

(b) Find the time taken for the van to travel from A to the midpoint of AB.

(5)

The van has mass 1200 kg.

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During its motion the van experiences a constant resistive force of magnitude 260 N

(c) Find the magnitude of the driving force exerted by the engine of the van as it travels from A to B.

(3)

(Total for question = 10 marks)

(**Q03 WME01/01, Jan 2024**)

Q5.

A car is moving at a constant speed of 25 m s^{-1} along a straight horizontal road.

The car is modelled as a particle.

At time $t = 0$, the car is at the point A and the driver sees a road sign 48 m ahead.

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Let t seconds be the time that elapses after the car passes A.

In a **first** model, the car is assumed to decelerate uniformly at 6 m s^{-2} from A until the car reaches the road sign.

(a) Use this first model to find the speed of the car as it reaches the sign.

(2)

The road sign indicates that the speed limit immediately after the sign is 13 m s^{-1}.

In a **second** model, the car is assumed to decelerate uniformly at 6 m s^{-2} from A until it reaches

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a speed of 13 m s^{-1}. The car then maintains this speed until it reaches the road sign.

(b) Use this second model to find the value of t at which the car reaches the sign.

(4)

In a **third** model, the car is assumed to move with constant speed 25 m s^{-1} from A until time t = 0.2, the car then decelerates uniformly at 6 m s^{-2} until it reaches a speed of 13 m s^{-1}. The car then maintains this speed until it reaches the road sign.

(c) Use this third model to find the value of t at which the car reaches the sign.

(4)

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(Total for question = 10 marks)

(Q03 WME01/01, Oct 2021)

Q6.



Figure 2

Three points P, Q and R are on a horizontal road where PQR is a straight line.

The point Q is between P and R, with $PQ = 6x$ metres and $QR = 5x$ metres, as shown in Figure 2.

A vehicle moves along the road from P to Q with constant acceleration.

The vehicle is modelled as a particle.

At time $t = 0$, the vehicle passes P with speed u m s^{-1}

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At time $t = 12$ s, the vehicle passes Q with speed $2u$ m s^{-1}

Using the model,

(a) show that $x = 3u$

(2)

As the vehicle passes Q, the acceleration of the vehicle changes instantaneously to 1.5 m s^{-2}

The vehicle continues to move with a constant acceleration of 1.5 m s^{-2} and passes R with speed $3u$ ms^{-1}

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Using the model,

(b) find the value of u,

(3)

(c) find the distance travelled by the vehicle during the first 14 seconds after passing P

(4)

(Total for question = 9 marks)

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(Q05 WME01/01, Jan 2023)

Q7.

A helicopter is hovering at rest above horizontal ground at the point H. A parachutist steps out of the helicopter and immediately falls vertically and freely under gravity from rest for 2.5 s. His parachute then opens and causes him to immediately decelerate at a constant rate of 3.9 m s^{-2} for T seconds ($T < 6$), until his speed is reduced to V m s^{-1}. He then moves with this constant speed V m s^{-1} until he hits the ground. While he is decelerating, he falls a distance of 73.75 m. The total time between the instant when he leaves H and the instant when he hits the ground is 20 s.

The parachutist is modelled as a particle.

(a) Find the speed of the parachutist at the instant when his parachute opens.

(1)

(b) Sketch a speed-time graph for the motion of the parachutist from the instant when he leaves H to the instant when he
hits the ground.

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(2)

(c) Find the value of T.

(5)

(d) Find, to the nearest metre, the height of the point H above the ground.

(4)

(Total for question = 12 marks)

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(Q07 WME01/01, Jan 2021)

Q8.

A small stone is projected vertically upwards with speed 20 m s^{-1} from a point O which is 5 m above horizontal ground. The stone is modelled as a particle moving freely under gravity.

Find

(a) the speed of the stone at the instant when it is 2 m above the ground,

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(2)

(b) the total time between the instant when the stone is projected from O and the instant when it first strikes the ground.

(4)

(Total for question = 6 marks)

(Q01 WME01/01, Jan 2021)

Q9.

Unless otherwise indicated, whenever a numerical value of g is required, take $g = 9.8$ m s^{-2} and give your answer to either 2 significant figures or 3 significant figures.

A car is moving along a straight horizontal road with constant acceleration a m s^{-2} ($a > 0$). At time $t = 0$ the car passes the point P moving with speed u m s^{-1}. In the next 4 s, the car travels 76 m and then in the following 6 s it travels a further 219 m.

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Find

(i) the value of u,

(ii) the value of a.

(Total for question = 7 marks)

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(**Q01 WME11/01, Specimen papers**)

Q10.

A car moves along a straight horizontal road with constant acceleration a ms $^{-2}$ where $a > 0$

The car is modelled as a particle.

At time $t = 0$, the car passes point A and is moving with speed u ms $^{-1}$

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In the first three seconds after passing A the car travels 20 m.

In the fourth second after passing A the car travels 10 m.

The speed of the car as it passes point B is 20 ms $^{-1}$

Find the time taken for the car to travel from A to B.

(Total for question = 8 marks)

(**Q02 WME01/01, June 2021**)

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